

MUHAMMAD HASHIR TARIQ

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PROFESSIONAL SUMMARY

Final-year Telecommunications Engineering student at NED University of Engineering & Technology with hands-on experience in wireless networks, RF systems, hardware programming, and AI-driven applications. Skilled in Python, Verilog, MATLAB, and 3D ray-tracing simulation tools. Currently conducting research on Reconfigurable Intelligent Surfaces (RIS) for 6G infrastructure. Seeking a Science & Engineering Associate role to leverage strong technical, analytical, and leadership capabilities in a project-driven environment.

EDUCATION

NED University of Engineering & Technology

Expected Graduation: August 2026

Bachelor of Engineering in Telecommunications

CGPA: 3.57 / 4.00

Relevant Coursework: Wireless Technologies & RF Planning, Digital Communications, Signals & Systems, Antenna Theory, Microwave Engineering, Computer Networks, Telecommunication Switching Systems.

DJ Sindh Science Govt College

Graduated

FSC Pre-Engineering

Karachi, Pakistan

PROFESSIONAL EXPERIENCE

Research Intern – IoT Labs | 6G Technology Research

Sept 2025 – October 2025 (8 Weeks)

NED University / IoT Labs

Karachi, Pakistan

- Conducted advanced research on Reconfigurable Intelligent Surfaces (RIS) as a core 6G enabling technology, focusing on programmable metasurface panels to control electromagnetic wave propagation.
- Modeled RIS-assisted channel environments using Wireless InSite (3D ray-tracing tool) to evaluate signal enhancement, energy efficiency, and spectral performance without active RF components.
- Formulated research documentation regarding strategic RIS deployment scenarios within urban 6G infrastructure layouts.

Engineering Intern

July 2025 – August 2025 (6 Weeks)

Pakistan Aeronautical Complex (PAC)

Kamra, Pakistan

- Analyzed large-scale telephony network architectures and telecommunication exchange operations to optimize data/voice transmission routing.
- Assisted in the configuration, testing, and deployment of analog, digital, VoIP, and IP-based telephone communication systems.
- Executed hands-on E1 cable handling, fiber optic splicing, and configured media converters to uplink high-speed internet services via LAN.
- Managed and provisioned Cisco and Juniper switches to maintain network security and stable connectivity across multiple remote zones.

LEADERSHIP & EXTRA-CURRICULAR EXPERIENCE

Vice President – TE LINKS Society

2025 – 2026

NED University

Karachi, Pakistan

- Directed a multidisciplinary student committee to bridge the gap between academia and the telecom industry through technical workshops, seminars, and networking events.
- Managed operational workflows, coordinated with faculty advisors, and oversaw marketing and student engagement strategies.

Event Manager – EVOLVE Gaming Event | NED Evolve '26 Festival

April 2026

NED University

Karachi, Pakistan

- Spearheaded the planning, logistics, and execution of the premier campus esports tournament, managing registration workflows and technical infrastructure for hundreds of participants.
- Curated and coordinated digital media assets, stream layouts, and technical network setups to ensure seamless, low-latency tournament operations.

TECHNICAL PROJECTS

RIS Implementation in 6G Infrastructure (Final Year Project)

- Designing and simulating an optimized Reconfigurable Intelligent Surface (RIS) system to maximize wireless coverage boundaries in 6G network layouts.
- Executing 3D ray-tracing propagation modeling via Wireless InSite software to measure beamforming capabilities and phase-shift optimizations.

FPGA Implementation – 5-bit Arithmetic Logic Unit (ALU)

- Developed and synthesized a 5-bit ALU using Verilog HDL, deploying the final architectural design onto a physical FPGA board to validate logical design constraints.

Laser & Optical Data Transmission System

- Prototyped a free-space optical communication system using laser hardware to wirelessly transmit and decode audio and data streams.
- Built hardware modulation/demodulation circuits to successfully isolate and recover signal inputs across the optical channel.

Wireless Picture & Image Transmission System

- Engineered a wireless visual data transmission system applying signal encoding, channel modeling, and error-handling techniques to optimize bandwidth efficiency.

TECHNICAL SKILLS

Programming: Python (Pandas, NumPy, Matplotlib, Scikit-learn), C/C++, Verilog HDL, HTML/CSS

Engineering Tools: MATLAB, Wireless InSite (3D Ray Tracing), Git/GitHub, Packet Tracer

Telecommunications: RF Propagation Modeling, Link Budget Analysis, Modulation Techniques (FDM/OFDM), LAN/WAN Setup, Fiber Splicing

Enterprise Software: Microsoft Office (Excel, Word, PowerPoint)

CERTIFICATIONS & COURSES

- Cisco Network Fundamentals (CCNA Track) – In Progress (Expected Completion: 2026)
- AI and Deep Learning – Harvard University (Coursera), 2024
- Python Programming – Google (Coursera), 2023
- AI Essentials – Google, 2024
- AI and Data Science – Saylani Mass IT Program (Completed)